

Pre-2000

2000-2010

2010-2020

2020-2025

Foundations

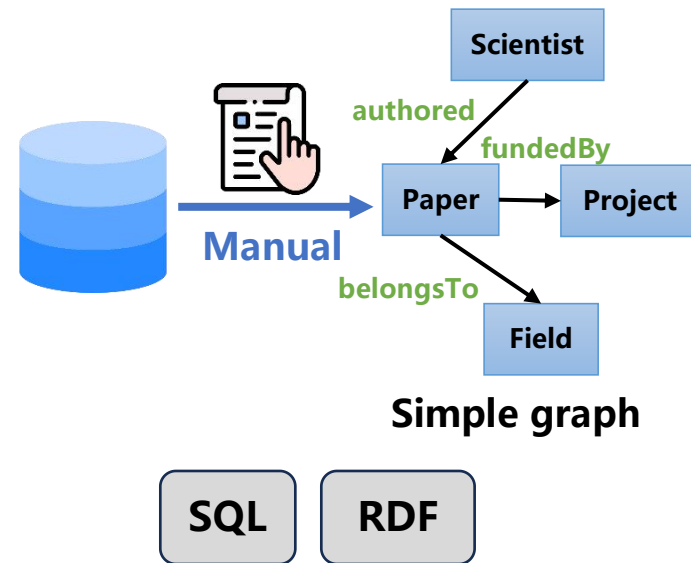
Integration & Semantics

Graph Learning

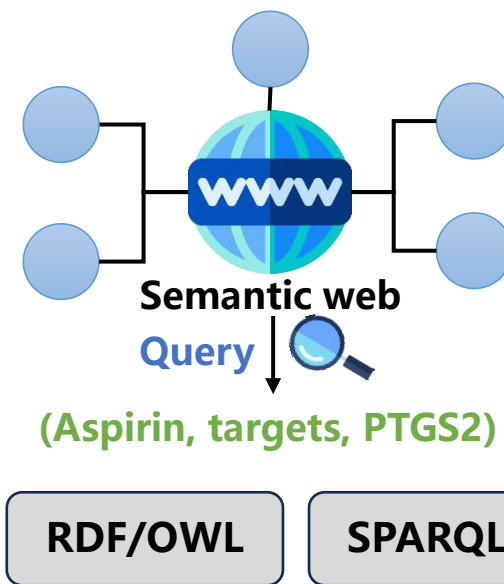
LLM-Coupled KGs

KG Technology Evolution

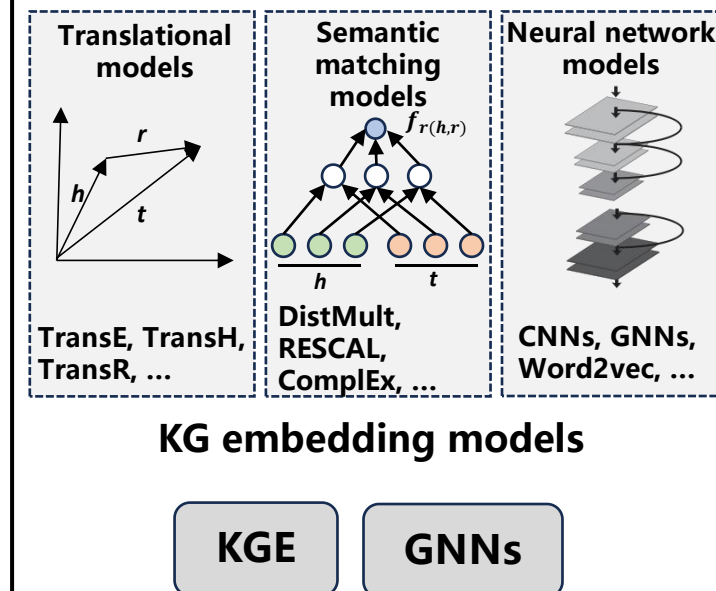
Relational databases; Static, hand-curated links



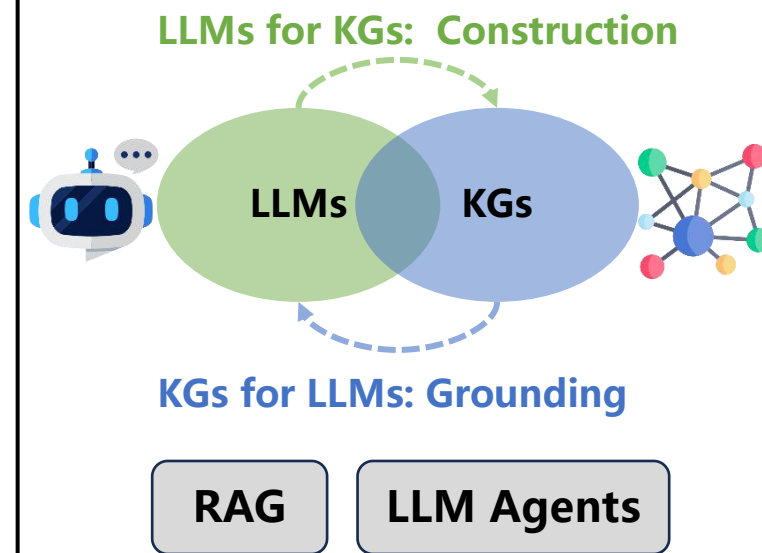
Semantic web tooling; Query over linked data



Graph representation learning



Bidirectional synergy: LLMs for KGs, and KGs for LLMs



Catalog & Query

Integrate & Analyze

Graph-Based Inference

LLM-Driven Discovery

KG in Scientific Practice

Ontology normalization

- ▶ MeSH (1960)
- ▶ SNOMED (1974)
- ▶ Unified Medical Language System (UMLS) (1991)
- ▶ GENE ONTOLOGY (1998)
- ▶ ...

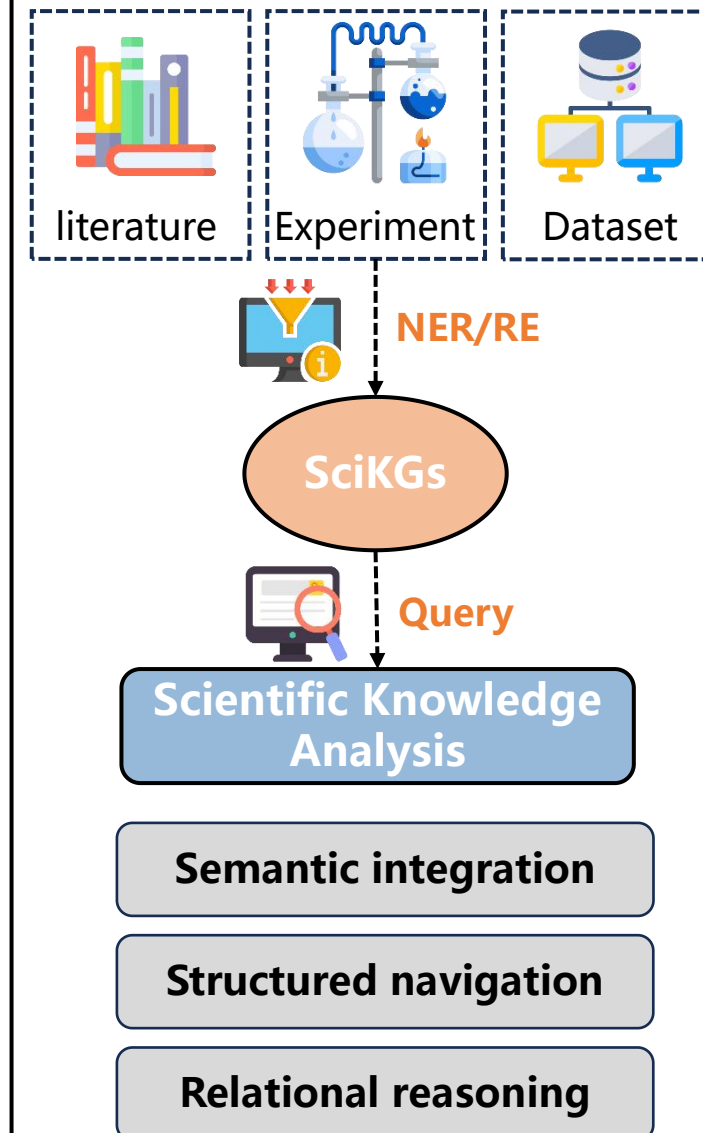
e.g., heart attack

Keyword/fielded search

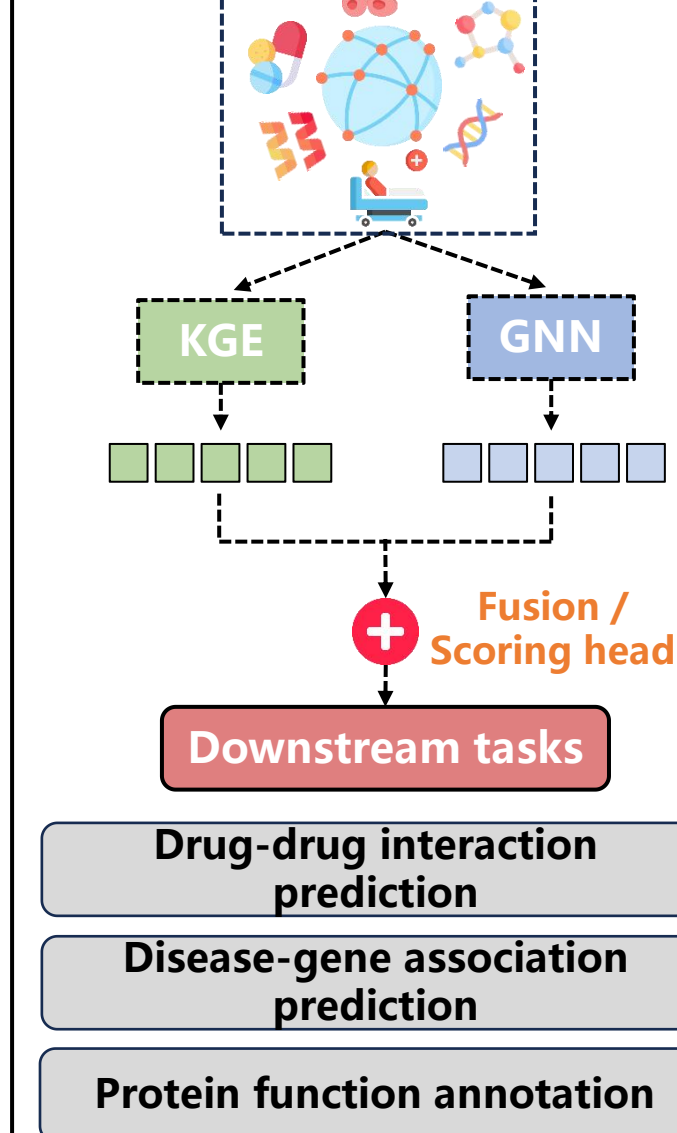
• MeSH: Myocardial Infarction (D009203)
Includes synonyms: heart attack, cardiac infarction, MI

Record lookup (only)

Multi-source datasets



SciKGs



SciKG-LLM integration

